

```
In [1]: import pandas as pd
```

```
In [2]: data = pd.read_csv("trends.csv")
df = pd.DataFrame(data)
```

```
In [3]: df.head()
```

```
Out[3]:
```

	location	year	category	rank	query
0	Global	2001	Consumer Brands	1	Nokia
1	Global	2001	Consumer Brands	2	Sony
2	Global	2001	Consumer Brands	3	BMW
3	Global	2001	Consumer Brands	4	Palm
4	Global	2001	Consumer Brands	5	Adobe

```
In [4]: df.tail()
```

```
Out[4]:
```

	location	year	category	rank	query
26950	Vietnam	2020	Là Gì?	1	Virus Corona là gì
26951	Vietnam	2020	Là Gì?	2	Miền thị thực là gì
26952	Vietnam	2020	Là Gì?	3	Đầu cắt moi là gì
26953	Vietnam	2020	Là Gì?	4	Bệnh bạch hầu là gì
26954	Vietnam	2020	Là Gì?	5	Đông Lào là gì

```
In [5]: df.shape
```

```
Out[5]: (26955, 5)
```

```
In [6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 26955 entries, 0 to 26954
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   location    26955 non-null  object
1   year        26955 non-null  int64
2   category    26955 non-null  object
3   rank        26955 non-null  int64
4   query       26955 non-null  object
dtypes: int64(2), object(3)
memory usage: 1.0+ MB
```

```
In [7]: df.describe
```

```
Out[7]: <bound method NDFrame.describe of
query
0      Global  2001  Consumer Brands  1      Nokia
1      Global  2001  Consumer Brands  2      Sony
2      Global  2001  Consumer Brands  3      BMW
3      Global  2001  Consumer Brands  4      Palm
4      Global  2001  Consumer Brands  5      Adobe
...      ...      ...      ...      ...
26950 Vietnam  2020      Là Gì?      1      Virus Corona là gì
26951 Vietnam  2020      Là Gì?      2      Miễn thị thực là gì
26952 Vietnam  2020      Là Gì?      3      Đầu cắt moi là gì
26953 Vietnam  2020      Là Gì?      4      Bệnh bạch hầu là gì
26954 Vietnam  2020      Là Gì?      5      Đông Lào là gì

[26955 rows x 5 columns]>
```

```
In [8]: df.columns
```

```
Out[8]: Index(['location', 'year', 'category', 'rank', 'query'], dtype='object')
```

```
In [9]: df.dtypes
```

```
Out[9]: location    object
year              int64
category          object
rank              int64
query             object
dtype: object
```

```
In [10]: df.isnull().sum()
```

```
Out[10]: location    0
year              0
category          0
rank              0
query             0
dtype: int64
```

```
In [11]: df.nunique()
```

```
Out[11]: location      83
year                20
category           2450
rank                 5
query             18431
dtype: int64
```

In this notebook, I did basic data analysis on the trends.csv dataset using Pandas.

I loaded the dataset using `pd.read_csv()` and checked the first and last five rows.

The dataset has 26,955 rows and 5 columns: location, year, category, rank, and query.

The data contains 2 numerical columns and 3 text columns. The years range from 2001 to 2020.

The rank values are between 1 and 5, with an average of 3.

There are 83 locations, 2,450 categories, and 18,431 different search queries.

There are no missing values in the dataset.

I also found 10 duplicate rows, which should be removed before further analysis